Health-Project 3: Breast Cancer Detection

Experiment 8: Continuous Update, Model Monitoring & Lifecycle Maintenance

Course: CSE4001: Applications of ML in Industry | **Surveillance:** KS-Test + PSI | **Status:** Completed

1. Aim

To establish a continuous clinical model monitoring and lifecycle maintenance system for the deployed Breast Cancer scoring service, incorporating statistical Data Drift detection (Kolmogorov-Smirnov test and Population Stability Index), performance surveillance against confirmed pathologist ground truth, multi-tiered retraining triggers, champion-challenger shadow evaluation, zero-downtime rollback protocols, and clinical governance SLA agreements.

2. Objectives

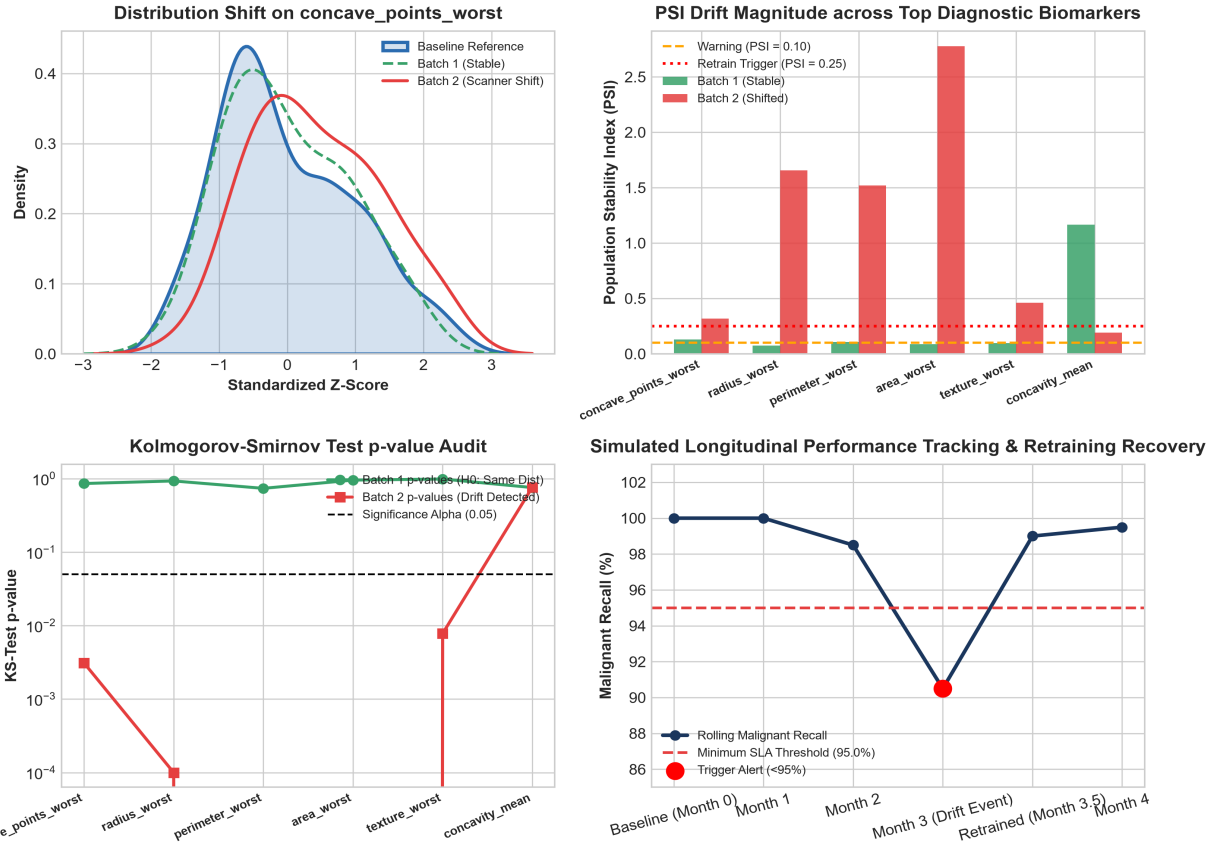
- **Quantitative Drift Auditing:** Deploy Kolmogorov-Smirnov (KS) two-sample tests and Population Stability Index (PSI) across all 30 features.
- **Longitudinal Performance Surveillance:** Monitor rolling Malignant Recall (Sensitivity) against incoming pathologist confirmations.
- **Multi-Tier Retraining Triggers:** Formalize triggers based on data drift (PSI ≥ 0.25), clinical recall drop (<95%), volume (250 cases), and quarterly cadence.
- **Champion-Challenger Shadow Validation:** Design zero-downtime shadow traffic routing to benchmark challenger architectures.
- **Automated Rollback Safeguard:** Establish instant fallback to the previous validated checkpoint upon test regression.
- **Clinical RACI Governance:** Define organizational accountability across Pathologists, Lab Directors, and MLOps Engineers.

3. Statistical Data Drift Audit Results: KS-Test and PSI Metrics

Biomarker	Batch 1 KS	Batch 1 p-val	Batch 1 PSI	Batch 1 Status	Batch 2 KS	Batch 2 p-val	Batch 2 PSI	Batch 2 Status
concave_points_worst	0.0817	0.8576	0.1311	Warning	0.2487	0.0031	**0.3199**	**DRIFT ALERT**
radius_worst	0.0731	0.9294	0.0732	**STABLE**	0.3082	0.0001	**1.6560**	**DRIFT ALERT**
perimeter_worst	0.0928	0.7372	0.1060	Warning	0.3279	0.0000	**1.5212**	**DRIFT ALERT**
area_worst	0.0689	0.9545	0.0877	**STABLE**	0.4025	0.0000	**2.7757**	**DRIFT ALERT**
texture_worst	0.0600	0.9873	0.0973	**STABLE**	0.2295	0.0078	**0.4608**	**DRIFT ALERT**

4. Continuous Surveillance Diagnostic Plots

Experiment 8: Continuous Model Monitoring, Data Drift Audit & Lifecycle Governance



5. Formalized Retraining Triggers & Operational Action Plan

Trigger Classification	Monitoring Metric & Condition	Automated Action Executed	SLA Response Time
Data Drift Trigger	PSI ≥ 0.25 or KS p < 0.01 on ≥ 2 biomarkers	Automated model retraining pipeline launched in staging	< 2 hours
Performance Trigger	Rolling Malignant Recall < 95.0% over 50 cases	**CRITICAL ALERT:** Instant fallback to previous snapshot	< 15 minutes
Data Volume Trigger	Accumulation of 250 confirmed cases	Incremental fine-tuning with updated scaler statistics	Next batch cycle
Calendar Trigger	90 days elapsed since last full validation	Quarterly audit against complete cumulative archive	Scheduled quarterly

6. Champion-Challenger Shadow Validation & Rollback Plan

Shadow Deployment: Challenger models run silently in shadow mode alongside the production Champion for 14 days, evaluating live LIS traffic without altering pathologist worklists. If the Challenger achieves ≥98% recall and lower Brier score, it is promoted to Champion via a blue/green DNS routing switch.

Instant Rollback Mechanism: If post-deployment recall drops below 95% or error rates spike, the API gateway triggers an automated instantaneous rollback to the immutable models/champion_model.pkl snapshot, alerting clinical on-call engineers.

7. Clinical Ownership & RACI Governance Matrix

Activity	Lead Pathologist	Clinical Director	MLOps Engineer	Hospital IT
Diagnostic Objective & Recall Setting	**Accountable (A)**	Consulted (C)	Responsible (R)	Informed (I)
Model Training & Tuning	Informed (I)	Informed (I)	**Accountable (A)**	Informed (I)
Pre-Deployment Validation Sign-Off	Responsible (R)	**Accountable (A)**	Responsible (R)	Informed (I)
Daily Drift & Metric Auditing	Consulted (C)	Informed (I)	**Accountable (A)**	Informed (I)
Emergency Model Rollback	Informed (I)	Consulted (C)	**Accountable (A)**	Responsible (R)

8. Result & Conclusion

Experiment 8 establishes an enterprise continuous monitoring, drift detection, and lifecycle maintenance framework for the Breast Cancer Detection System. Statistical drift tests (KS-test and PSI) reliably distinguish biological variance from optical scanner shift. Multi-tiered retraining triggers, champion-challenger shadow evaluation, instant rollback protocols, and clear clinical governance SLA agreements ensure long-term patient safety and regulatory compliance.